INCH-POUND

MIL-PRF-8565/17(AS)
09 February 2007
SUPERSEDING
MS17334P
22 April 1982
(See 6.9)

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 1/3-AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

REQUIREMENTS

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be D8565/17-1.

2. Dimensions and weight. The dimensions and the configuration of the battery container shall be as shown on figure 1 of this specification sheet. The weight of the battery shall be not greater than 3.5 pounds (1.59 kg).

3. Capacity. The rated capacity of the battery shall be 1/3-ampere-hours, based on its ability to conform to the life test final discharge. The life test final discharge is a 1-hour discharge at 24 °C ambient temperature with a 0.33-ampere load (1C-rate) to a cutoff voltage of 19 volts (0.33 Amp/1.0 hr/24 °C/19.0V). See 3.8.6 of MIL-PRF-8565.

4. Application. The battery shall not be designed for engine starting applications.

5. Mounting. For installation in the aircraft, the battery shall be mounted as shown on figure 3 of MIL-PRF-8565.

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6. Service life. The battery's service life (useful life) in the aircraft shall be not less than 3 years (36 months). The battery shall require no scheduled maintenance during its service life (see 3.8.9 and 6.15.10 of MIL-PRF-8565; see 4.5.10 and 6.11.7.2 as specified herein).

7. Qualification inspection and conformance inspection. Table I of this specification sheet lists the required inspections.

8. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

8.1 The following paragraphs are not applicable:

3.5.4 Venting.
3.5.10 Heaters.
3.6.5 Marking for batteries with heaters.
3.8.4 Dielectric strength and insulation resistance.
3.8.5 Vent tubes.
3.8.7 Constant voltage discharge.
3.8.8 Strength of receptacle.
3.8.12 Battery gas emission.
3.8.14 Impact resistance (nonmetallic battery container only).
3.8.16 Output performance.
3.8.17 Cold and hot temperature charge.
4.5.4 Dielectric strength and insulation resistance test.
4.5.5 Strength of vent tubes test.
4.5.8 Constant voltage discharge.
4.5.9 Strength of receptacle test.
4.5.18 Battery gas emission test.
4.5.24 Impact resistance test (nonmetallic battery container).
4.5.28 Output performance test.
4.5.29 Cold and hot temperature charge test.
4.5.30 Wormgear retaining pin hardness.

8.2 Modify the following paragraphs:

3.5.3 Battery container and cover. In the second sentence, delete “hold-downs, vent tubes”.

3.5.5 Receptacles. Delete the second sentence and substitute: “The battery shall contain an electrical connector dimensionally equal to MS3102R-14S-7S. The connector shall be installed on the battery in the location shown on figure 1 of MIL-PRF-8565/17(AS), and the key of the connector shall be in the 12 o’clock position. The connector socket pin allocations shall be in accordance with the following:

“a. Socket A shall be battery negative (-) for battery charging and discharging.

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"b. Socket B shall be +28 VDC input for battery charging and positive (+) for battery discharge output.

"c. Socket C shall not be used (spare) in this application. The battery shall be designed so that no internal components of the battery make mechanical or electrical contact with socket C."

3.5.7 Electrolyte leakage. Delete "except when tested in accordance with 4.5.18".

3.6.3 Identification marking. In the third sentence, delete "figure 1 herein" and substitute "figure 2 of MIL-PRF-8565/17(AS)".

3.6.4 Battery caution marking. In the first sentence, delete "figure 2 herein" and substitute "figure 3 of MIL-PRF-8565/17(AS)".

3.7 General requirements. Delete the text of 3.7.e and substitute: "Electrolyte leakage or spilling of electrolyte at any time during the test. Corrosive or liquid discharge shall not occur during testing to this specification. Products of cell venting that are discharged shall be non-toxic."

8.3 Add the following paragraph to MIL-PRF-8565:

"3.8.18 Pulsed high rate discharge. The battery potential shall be not less than 10.6 volts throughout each of the three discharges after being tested in accordance with 4.5.31, as specified in MIL-PRF-8565/17(AS)."

VERIFICATION

1. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows:

1.1 Modify the following MIL-PRF-8565 paragraphs:

4.3.4 Group A inspection. Delete the last sentence.

4.4.2 Constant potential charging method. In the first sentence, delete "28.25 ± 0.25 volts for 2.0 ± 0.1 hours" and substitute "28.5 ± 0.5 volts for 5.0 ± 0.1 hours".

4.4.3 Constant current charging method. Delete "C/10 (± 3 percent)" and substitute "0.075 amperes".

4.5.6 Electrical test preparation. In the first sentence, delete "18.0 volts" and substitute "19.0 volts".

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4.5.7 Capacity discharge. In the second sentence, delete “18.0 volts” and substitute “19.0 volts”.

4.5.10 Cycling test. Delete the text of 4.5.10 and substitute: “Constant potential charge the battery in accordance with 4.4.2, as modified by MIL-PRF-8565/17(AS), at room temperature (see 4.4.1 of MIL-PRF-8565). Subject the battery to 100 cycles of discharge and charge under the ambient conditions of 4.4.1 of MIL-PRF-8565. Each cycle shall consist of a 1.0-hour, 1.0C-rate discharge, followed by a 5.0-hour constant potential charge of 28.5 volts, followed by a 1-hour rest period. If the battery is unable to meet requirement (1) of table III as modified by MIL-PRF-8565/17(AS) for three consecutive discharges, one reconditioning is permitted and the cycling test shall continue. The battery shall be reconditioned as follows:

“a. Discharge the battery at the 1.0C-rate to 19.0 volts.

“b. Constant current charge the battery at 0.075 ampere for 16.0 ± 0.1 hours.

“c. Discharge the battery at the 1.0C-rate to 19.0 volts.

“d. Constant current charge the battery at 0.075 ampere for 16.0 ± 0.1 hours.

“e. Examine the battery for compliance with 3.8.9 of MIL-PRF-8565.”

4.5.11 Capacity discharge at 49 °C (120 °F). In the third sentence, delete “18.0 volts” and substitute “19.0 volts”.

4.5.12 Charge and discharge test at low temperature. In steps c and f (second sentence), Delete “18.0 volts” and substitute “19.0 volts”.

4.5.13 Discharge while inverted. In step d, delete “requirement (5)” and substitute “requirement (4)”.

4.5.14 Altitude test. In steps c through h, delete the text and substitute:

“c. Within 15 minutes after completing step 4.5.14b of MIL-PRF-8565, lower the chamber temperature to 29 °C (-20 °F) and lower pressure inside chamber to simulate ambient pressure at temperature environment.

“d. Discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever comes first. Examine the battery for the requirements of 3.8.10 of MIL-PRF-8565.

“e. Constant potential charge the battery in accordance with 4.4.2, as modified by MIL-PRF-8565/17(AS) for 2 hours.

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"f. Open-circuit the battery and, within 5 minutes, return the chamber to ground ambient air pressure (see 4.4.1 of MIL-PRF-8565).

"g. Discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever comes first. Examine the battery for the requirements of 3.8.10 of MIL-PRF-8565.

"h. Examine the battery for the requirements of 3.7, as modified by MIL-PRF-8565/17(AS) and 3.5.9 of MIL-PRF-8565."

4.5.15 Shock test (basic design). In step f, delete the sentence and substitute: "After the fifteenth impact, discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever occurs first."

4.5.16 Temperature shock test. In step b, delete the second sentence. In step d, delete the second sentence. In step f, delete the second sentence. In step j, delete the text and substitute "discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever comes first".

4.5.17 Temperature rise and float test. In step e, delete "18.0 volts" and substitute "19.0 volts".

4.5.19 Vibration test. In step c, insert the following after the first sentence: "Vibration isolators shall not be used". In step g, delete the text and substitute: "Discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever occurs first."

4.5.20 Humidity test. In step i, delete the text and substitute: "Discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever occurs first."

4.5.21 Salt fog test. In step j, delete the text and substitute: "Discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever occurs first."

4.5.22 Ground storage tests. In steps d and i, delete "18.0 volts" and substitute "19.0 volts".

4.5.23 Physical integrity at high temperature test. In step b, delete "18.0 volts" and substitute "19.0 volts". In step e, delete the last sentence. In step k, delete "18.0 volts" and substitute "19.0 volts".

4.5.25 Shock test (crash safety). In step f, delete the last sentence and substitute: "Discharge the battery at the 6.5C-rate for 3 minutes or to a cutoff voltage of 15.0 volts, whichever occurs first."

4.5.26 Shelf life test. After the third sentence, delete the text and substitute: "Constant potential charge the battery in accordance with 4.4.2, as modified by MIL-PRF-8565/17(AS). Stabilize the battery at room temperature (see 4.4.1 of MIL-PRF-8565). Discharge the battery at

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the C-rate to 19.0 volts. Constant potential charge the battery in accordance with 4.4.2, as modified by MIL-PRF-8565/17(AS). Subject the battery to the cycling test of 4.5.10, as modified by MIL-PRF-8565/17(AS). Examine the battery for the requirements of 3.8.9 and 3.8.13.2, both of MIL-PRF-8565."

4.5.27 Deep discharge recovery test. In step b, delete "1.0 ohms" and substitute "5.0 ohms". In step d, delete "C/10 (± 3 percent)" and substitute "0.075 ampere". In step f, delete "C/10 (± 3 percent)" and substitute "0.075 ampere".

1.2 MIL-PRF-8565, paragraph 4.5.31 delete and substitute:

"4.5.31 Pulsed high rate discharge test. Constant potential charge the battery in accordance with 4.4.2, as modified by MIL-PRF-8565/17(AS). Discharge the battery at 9.12 amperes for 20 seconds, followed by a 2-minute rest period. Repeat the discharge/rest period cycle two additional times. Examine the battery for compliance with 3.8.18, as specified in MIL-PRF-8565/17(AS)."

1.3 Add the following paragraph to MIL-PRF-8565:

4.5.32 Final examination. Without dissecting the battery, examine it for the requirements of 3.7, as modified by MIL-PRF-8565/17(AS).

NOTES

1. Modify the following paragraph.

6.9 Cross-reference. Insert the following after the first sentence: "This specification supersedes MS17334P(AS) in its entirety. PIN D8565/17-1 supersedes both PINs MS17334-1 and MS17334-2."

2. Add the following paragraphs to MIL-PRF-8565.

"6.11.7 Additional application information. The following information is provided for users of the D8565/17-1 battery.

"6.11.7.1 Initial installation date. The initial installer should write the date the battery is first installed in an aircraft (date first placed in service) in the space provided on the battery's identification marking (see figure 2 of MIL-PRF-8565/17(AS)).

"6.11.7.2 Service life and removal from aircraft. The battery's service life (useful life) in the aircraft is limited to 36 months. The battery should be removed from aircraft service 36 months after the date it was first installed in an aircraft (see requirement 6 of MIL-PRF-8565/17(AS)).

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"6.11.8 Environmental characteristics. Battery performance should not be adversely affected when operating in a sand and/or dust environment. Battery materials should not support fungus growth. The battery should operate safely in an atmosphere containing ignitable or explosive fumes."

TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1. (G)	4.5.1 (G)		X	X
2	VISUAL AND MECHANICAL EXAMINATION 2/	X	X	X	2/ 3.5 2/ 3.6 , 3.8.2 (G), table II (G)	4.5.2 (G)	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2 (N), 3.8.3 (G)	4.5.3 (G)	X 3/	X	
4	ELECTRICAL TEST PREPARATION	X	X	X		4.5.6 (S)		X	X
5	CAPACITY DISCHARGE	X	X		Reqmt 3 (N), 3.8.6 (G)	4.5.7 (S)	X	X	X
6	CYCLING TEST	X			Reqmt 6 (N), 3.8.9 (G)	4.5.10 (S)		X	
7	CAPACITY DISCHARGE AT 49 °C		X		3.8.6 (G), reqmt (3) of table III (S)	4.5.11 (S)			
8	CHARGE AND DISCHARGE TEST AT LOW TEMPERATURE			X	3.8.6 (G), reqmt (2) of table III (S)	4.5.12 (S)			X
9	DISCHARGE WHILE INVERTED		X		3.5.9 (G), reqmt 4 of table III (S)	4.5.13 (S)			
10	ALTITUDE TEST		X		3.5.9 (G), 3.7 (S), 3.8.10 (G)	4.5.14 (S)			X
11	SHOCK TEST (BASIC DESIGN)	X			3.7 (S), 3.8.10 (G)	4.5.15 (S)			
12	TEMPERATURE SHOCK TEST	X			3.7 (S), 3.8.10 (G)	4.5.16 (S)		X	
13	TEMPERATURE RISE AND FLOAT TEST		X		3.7 (S), 3.8.11 (G), reqmt (1) of table III (S)	4.5.17 (S)			X
14	VIBRATION TEST		X		3.7 (S), 3.8.10 (G)	4.5.19 (S)			X

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TABLE I. Qualification inspection and conformance inspection of batteries. - Continued.

Test Number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
15	HUMIDITY TEST		X		3.7 (S), 3.8.10 (G)	4.5.20 (S)		X	X
16	SALT FOG TEST	X			3.7 (S), 3.8.10 (G)	4.5.21 (S)			
17	SHOCK TEST (CRASH SAFETY)	X			3.8.10 (G), 3.8.15 (G)	4.5.25 (S)			
18	GROUND STORAGE TEST			X	3.8.13.1 (G)	4.5.22 (S)			
19	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7. (S), 3.8.10 (G), reqmt (1) of table III (S)	4.5.23 (S)			
20	DEEP DISCHARGE RECOVERY TEST			X	3.7 (S), 3.8.13.3 (G)	4.5.27 (S)			
21	SHELF LIFE TEST			X	3.8.9 (G), 3.8.13.2 (G) reqmt (1) of table III (S)	4.5.26 (S)			
22	PULSED HIGH RATE DISCHARGE TEST		X		3.8.18 (N)	4.5.31 (N)		X	
23	FINAL EXAMINATION	X	X	X	3.7 (S)	4.5.32 (S)		X	X

1/ The notations in parentheses specify the source of the paragraph or table. "(G)" means the paragraph is as shown in MIL-PRF-8565. "(S)" shows that the paragraph or table has been modified by this specification sheet. "(N)" indicates that a new paragraph or requirement has been added by this specification sheet.

2/ Visual examination is required for requirements 3.5 through 3.5.10. Color and marking examination is required for requirements 3.6 through 3.6.6. Requirements 3.5.3, 3.5.5, 3.5.7, 3.6.3, and 3.6.4 are as modified by this specification sheet. Requirements 3.5, 3.6 and all other 3.5 and 3.6 subparagraphs are as specified in MIL-PRF-8565.

3/ See sample information in test paragraph.

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TABLE III: Delete and substitute:

TABLE III. Capacity performance requirements.

Requirement number	Discharge cutoff voltage and test temperature	Discharge rate factor	Minimum time to cutoff voltage
(1)	19.0 volts at room temperature	1.0C	60 minutes
(2)	19.0 volts at 49 ± 2 ºC	1.0C	66 minutes
(3)	19.0 volts at -18 ± 2 ºC	1.0C	30 minutes
(4)	14.0 volts at room temperature	6.5C	5 minutes

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NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED TOLERANCES ARE ±.065.
3. ABBREVIATIONS USED IN THIS FIGURE ARE DIAMETER(ø), MAXIMUM(MAX), AND MINIMUM(MIN).

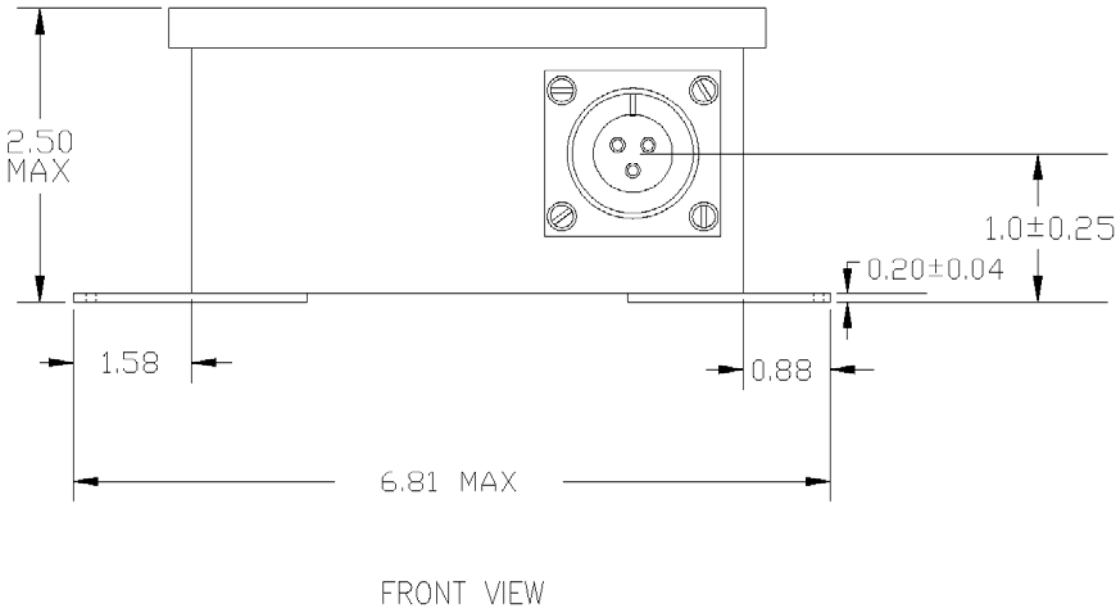


FIGURE 1. PIN D8565/17-1 dimensions.

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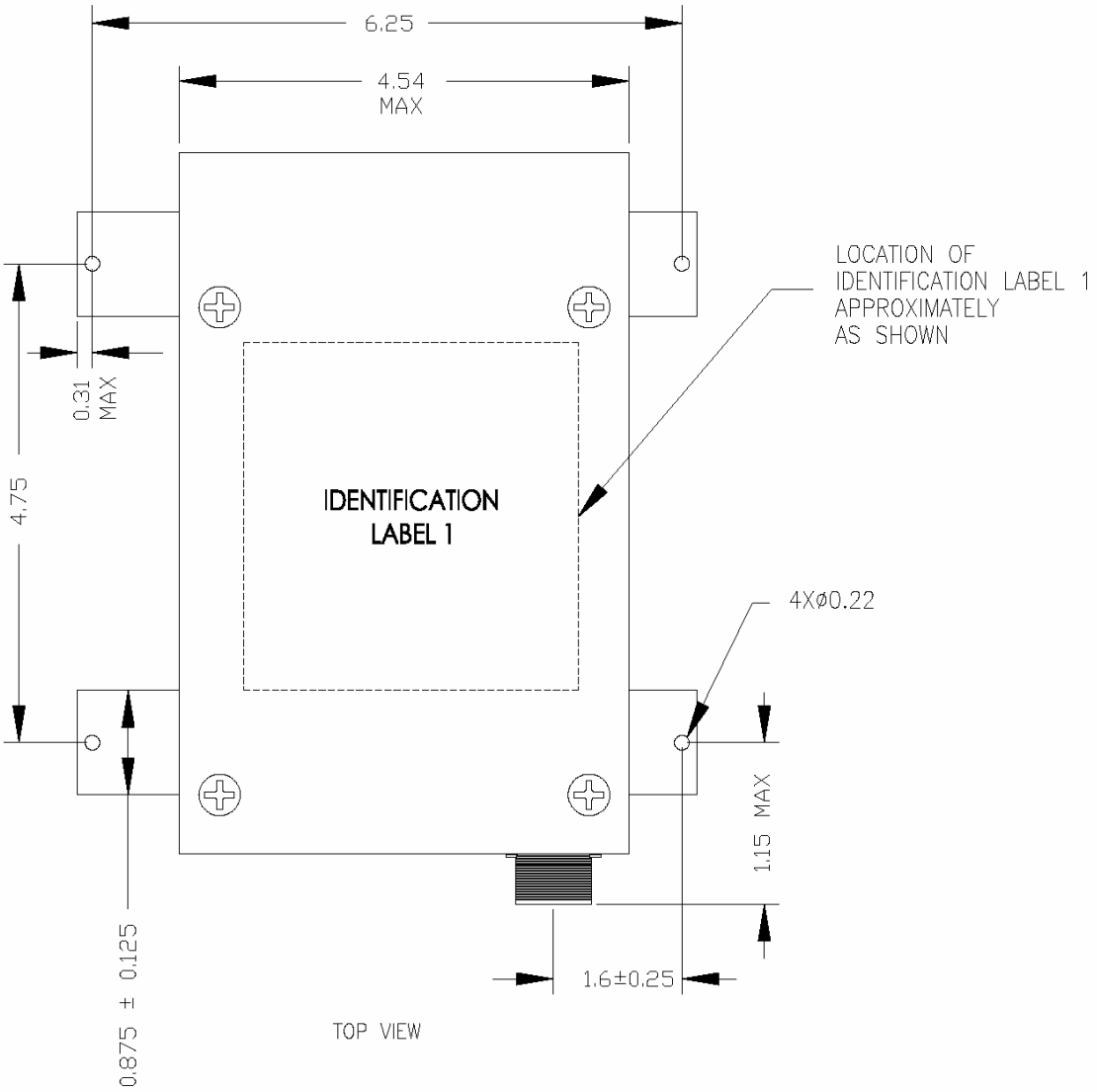


FIGURE 1. PIN D8565/17-1 dimensions – Continued.

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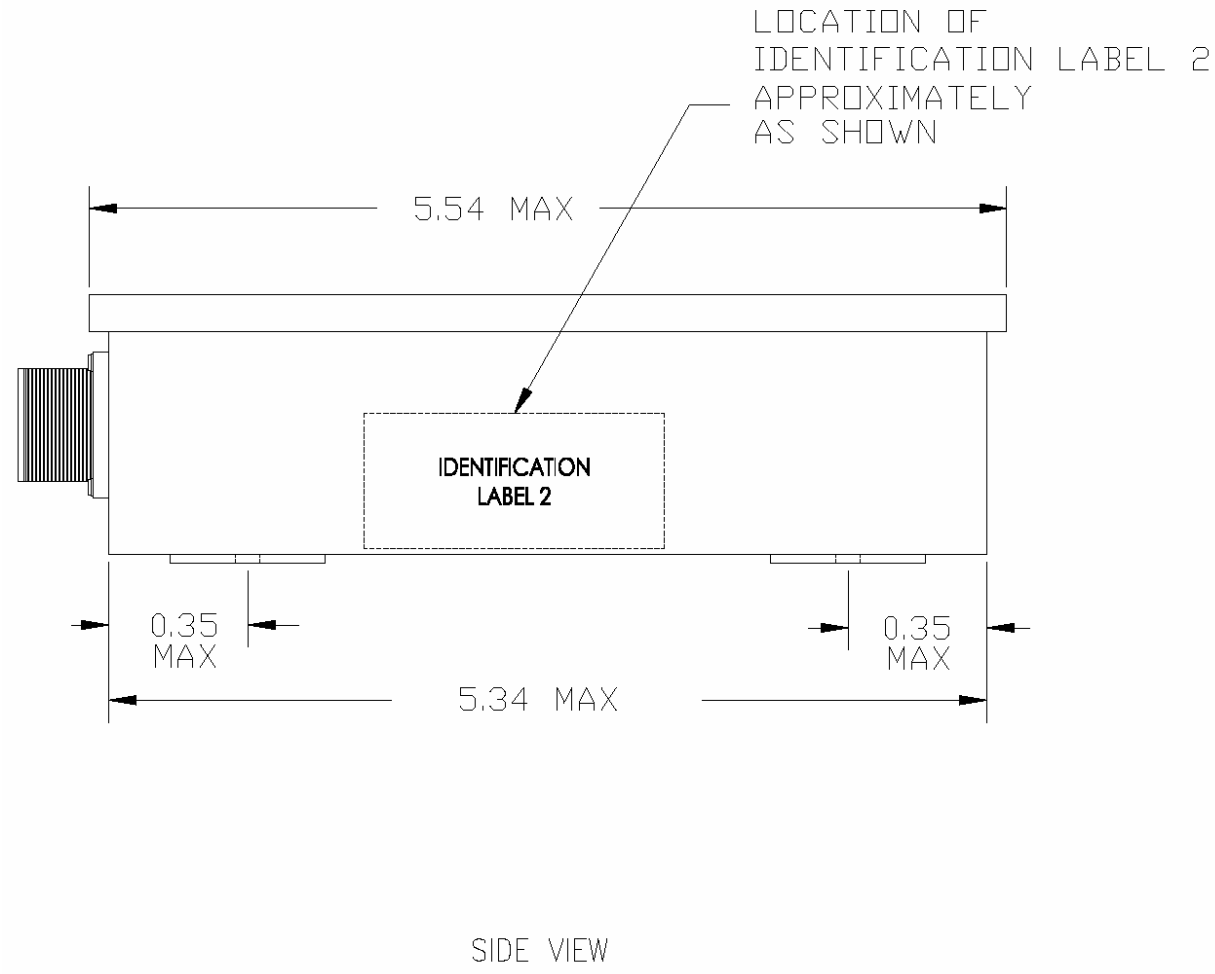


FIGURE 1. PIN D8565/17-1 dimensions - Continued.

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NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT 24-VOLT, MAINTENANCE-FREE
MILITARY PIN:	D8565/17-1
MAXIMUM WEIGHT:	3.5 LB.
CAPACITY:	0.33 AH (0.33 AMPS/1.0 HOUR/24 °C /19 V)
NATIONAL STOCK NUMBER:	6140-**-***-****
CONTRACT NUMBER:	***
LOT CODE:	***
DATE OF MANUFACTURE	***
BATTERY SERIAL NUMBER:	***
MANUFACTURER NAME:	***
DATE FIRST PLACED IN SERVICE:	

FIGURE 2. Marking for D8565/17-1 battery.

MAINTENANCE-FREE BATTERY

THIS BATTERY SHALL BE SERVICED
ONLY IN AN AUTHORIZED BATTERY SHOP
IN ACCORDANCE WITH
NAVAIR 17-15BAD-1

FIGURE 3. Battery caution marking.

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS

Agent:
Navy - NW
(Project 6140-2006-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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